Lecture 1

Production Possibility Frontier & Gains from Trade

ECO 101: Introduction to Microeconomics

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Roadmap

Lectures:

1. **Production Possibility Frontier & Gains from Trade**
2. Law of Demand and Supply
3. Elasticity
4. Government Actions in Markets
5. The Production Process: inputs & outputs
6. Perfect Competition
7. Monopoly
8. Game Theory

What is Economics?

Economics is the study of how people make choices when resources are scarce.

The fundamental problem: We have unlimited wants but limited resources, so we must make choices about how to use those resources.

Two main branches of economics:

1. **Microeconomics**: studies behaviour of individual economic players (consumers, producers, firms)
2. **Macroeconomics**: studies how an entire economy performs with regards to production and distribution of goods and services

Scarcity, Choice & Incentive

Scarcity
We have **unlimited wants but limited resources**.
→ We cannot have everything we want.

Choice
Because resources are scarce, we must **choose how to use them**.
→ Every choice means giving something else up.

Incentives
People respond to **rewards and costs** when making choices.
→ Changes in incentives can change behaviour.

For 3 hours tonight, you can study for ECO101, watch Netflix, or go out with friends.
You can't do all three → **scarcity**
You decide to study → **choice**
An upcoming exam gives you a reason to study → **incentive**

Opportunity Cost

Definition: The opportunity cost of an action is the **highest-valued alternative forgone.**

Example

On Friday, you can study for ECO101 exam, go to a friend's birthday party or earn Tk. 1,000 by tutoring. You decide to tutor.

- Your opportunity cost is **going to your friend's birthday party** *if that is the highest-valued alternative you would otherwise have chosen.*
- The opportunity cost is NOT the friend's birthday party and studying for ECO101 exam. It's the **highest-valued alternative forgone**.
- Opportunity cost can be 0-price. The Tk. 1,000 is the **monetary benefit of your choice**, not automatically its opportunity cost.
- Opportunity cost is subjective. It is NOT the same for everyone because everybody has different preferences.

Opportunity Cost

From choices to production

The quantities of goods and services that we can produce are limited by both our available resources and technology. If we want to increase our production of one good, we must decrease our production of another good. We face a **trade-off**.





Production Possibilities Frontier (PPF)

The production possibilities frontier (PPF) is the boundary between the combinations of goods and services that can be produced and those that cannot.

We focus on two goods for an efficient illustration. Assume an economy in which everything remains the same (*ceteris paribus*) except for the production of these two goods (coca colas & pizzas).

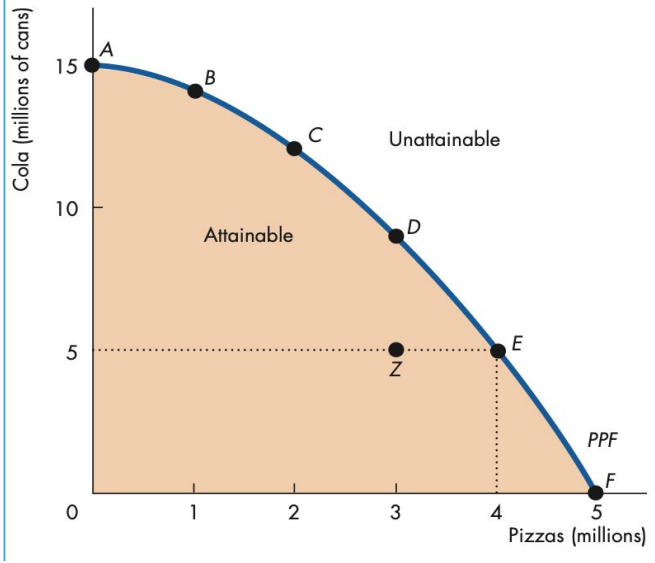
On the PPF: efficient allocation

Inside the PPF: inefficient allocation

Outside the PPF: unattainable with current resources

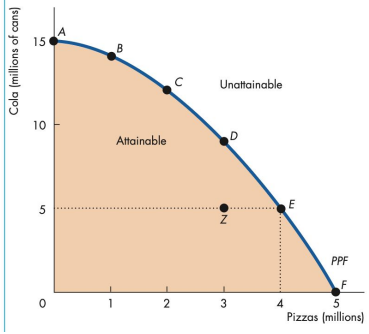
Along the PPF: a trade-off

Figure 2.1 The Production Possibilities Frontier



Opportunity Cost on the PPF

Figure 2.1 The Production Possibilities Frontier



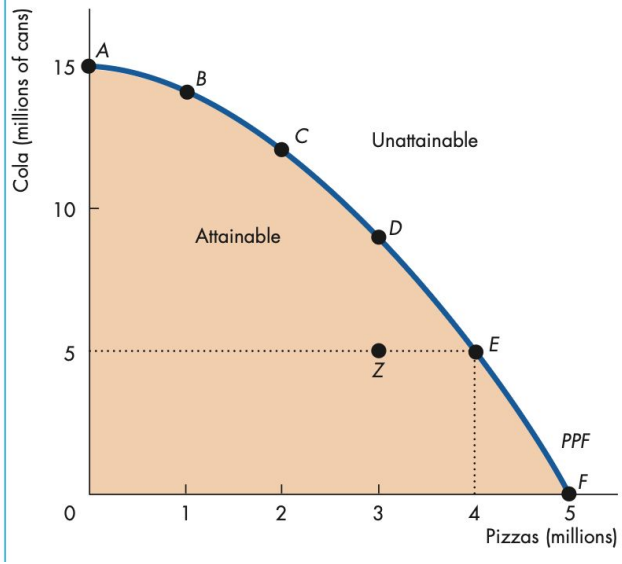
Possibility	Pizzas (millions)		Cola (millions of cans)
A	0	and	15
B	1	and	14
C	2	and	12
D	3	and	9
E	4	and	5
F	5	and	0

- At point A, no pizzas are being produced. All the resources are being expended to produce cola—which gives us 15 million Colas and 0 pizzas. Similarly, at point F, all the resources are allocated to produce pizzas. There are 5 million pizzas and 0 Cola.

- Z is an inefficient point inside the PPF, indicating that the resources are either wasted, misallocated, or both. At this point, it would be possible to use the available resources to produce more of either or both goods.
  Resources are wasted or unused when they are idle but could be working. E.g. idle factories.

Opportunity Cost on the PPF

Figure 2.1 The Production Possibilities Frontier



- Every choice along the PPF is a trade-off. So, we must give up something to get something else. All trade-offs involve an opportunity cost.
- What happens if we move from point D to point C?
- If we choose point C over point D, the additional 3 million cans of Cola cost 1 million pizzas. Similarly, the opportunity cost of 1 million Cola is 1/3 of pizzas when we move from point D to point C. Therefore, **opportunity cost is a ratio**.
- We achieve **production efficiency** if we produce goods and services at the lowest possible cost. This outcome occurs at all points on the PPF.

What shifts the PPF?

- More resources (workers, land, capital)
- Better technology (higher productivity)
- Natural disaster (reduces productivity/capacity)

Economic growth expands what an economy can produce.

Economic growth increases our standard of living, but it does not overcome scarcity or avoid opportunity cost. To make our economy grow, we face a trade-off – the faster we make production grow, the greater is the opportunity cost of economic growth.

The Cost of Economic Growth

Two factors that influence economic growth:

1. Technological Change: The development of new goods and services as well as better ways of producing goods and services.

2. Capital Accumulation: The growth of capital resources, including human capital.

New technologies and new capital have respective opportunity costs i.e., to use resources in research and development and to produce new capital, we must decrease our production of consumption goods and services.

PPF and Economic Growth

Economics in Action

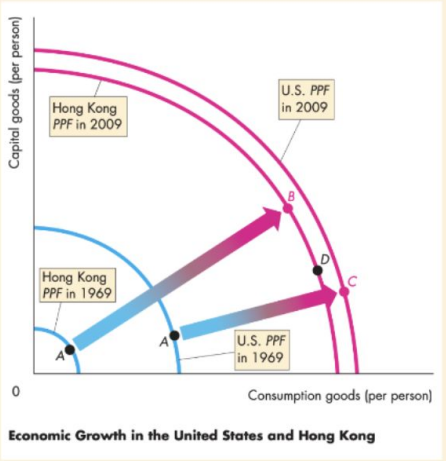
Hong Kong Catching Up to the United States

In 1969, the production possibilities per person in the United States were more than four times those in Hong Kong (see the figure). The United States devotes one fifth of its resources to accumulating capital and in 1969 was at point *A* on its *PPF*. Hong Kong devotes one third of its resources to accumulating capital and in 1969, Hong Kong was at point *A* on its *PPF*.

Since 1969, both countries have experienced economic growth, but because Hong Kong devotes a bigger fraction of its resources to accumulating capital, its production possibilities have expanded more quickly.

By 2009, production possibilities per person in Hong Kong had reached 94 percent of those in the United States. If Hong Kong continues to devote more resources to accumulating capital than we do (at point *B* on its 2009 *PPF*), it will continue to grow more rapidly. But if Hong Kong decreases capital accumulation (moving to point *D* on its 2009 *PPF*), then its rate of economic growth will slow.

Hong Kong is typical of the fast-growing Asian economies, which include Taiwan, Thailand, South Korea, China, and India. Production possibilities expand in these countries by between 5 and almost 10 percent a year.



Economic Growth in the United States and Hong Kong

If such high economic growth rates are maintained, these other Asian countries will continue to close the gap between themselves and the United States, as Hong Kong is doing.

More capital goods can lead to higher production of consumer goods and growth in the economy even though it might initially incur an opportunity cost.

Gains from Trade

People can produce for themselves all the goods and services that they consume, or they can produce one good or a few goods and trade with others. Producing only one good or a few goods is called **specialisation**. Trade can make both sides better.





Absolute Advantage vs. Comparative Advantage

Absolute Advantage

Can produce more with the same resources.

Such advantage arises from differences in productivity. Better technology, more capital, and a better skill set—all give a person/firm/nation an absolute advantage.

If you have an absolute advantage, you do not have a comparative advantage in every activity. It does not directly depend on opportunity cost.

Comparative Advantage

Can perform the activity at a lower opportunity cost than anyone else.

With specialisation and exchange, we can produce more and consume more than if we tried to make everything ourselves.

Absolute advantage involves comparing productivities – production per hour – whereas comparative advantage involves comparing opportunity cost.

Absolute Advantage vs. Comparative Advantage: Example

China 🇨🇳	
Cars	Clothes
50	0
25	50
0	100

China has an absolute advantage in producing both.

To understand how trade takes place in such a scenario, we need to calculate the opportunity cost for both of these countries when producing Clothes.

For 100 units of Clothes, China foregoes 50 units of Cars. Hence, the OC of Chinese production of 1 unit of Cloth = 50/100 units of Cars = ½ of a unit of Cars.

Similarly, the OC of China producing 1 unit of Car = 2 units of Clothes.

Bangladesh 🇧🇩	
Cars	Clothes
10	0
5	20
0	40

For 40 units of Clothes, Bangladesh foregoes 10 units of Cars.

Hence, the OC of Bangladesh producing 1 unit of Cloth = 10/40 or ¼ of a unit of Car.

Similarly, the OC of Bangladesh producing 1 unit of Cars = 4 units of Clothes.

Clearly, the OC of producing Cars is less in China than in Bangladesh. Therefore, China should specialise in producing Cars.

When do we have no trade?

When the opportunity cost for producing a good is the same for two producers, neither has a comparative advantage.

In that case, there would be no benefit from specialisation or trade since both the countries are equally efficient at producing both the goods.

No difference in opportunity cost. No Comparative advantage. No trade

No Gains from Trade: Example

Suppose **both countries have exactly the same opportunity costs.** In that case, neither has a comparative advantage, so specialisation cannot create gains from trade.

Country	Rice (tonnes)	Coffee (tonnes)
Bangladesh	20	10
Vietnam	10	5

For Bangladesh,

- 20 tonnes Rice = 10 tonnes Coffee foregone
- 1 tonne Rice = 0.5 tonne Coffee foregone
- 1 tonne Coffee = 2 tonnes Rice foregone

For Vietnam,

- 10 tonnes Rice = 5 tonnes Coffee foregone
- 1 tonne Rice = 0.5 tonne Coffee foregone
- 1 tonne Coffee = 2 tonnes Rice foregone

Exit ticket

Before you leave the class today, answer the following:

1. What is an opportunity cost of one choice you made today?

2. Why is a PPF usually bowed outward rather than a straight line?

3. How can two countries gain from trade?

4. When can there be no gains from trade?